ME 701

Gear Profile Optimization

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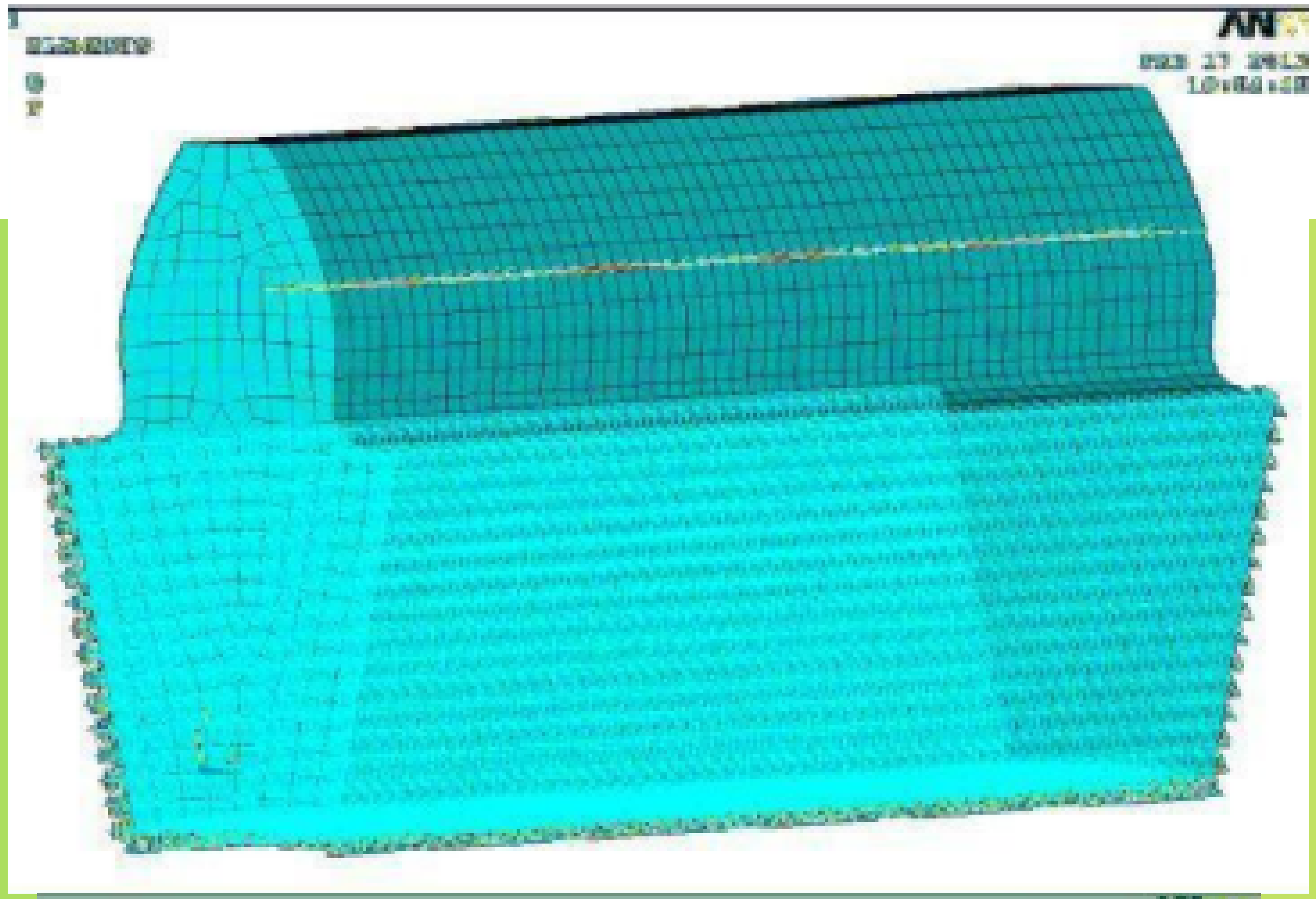
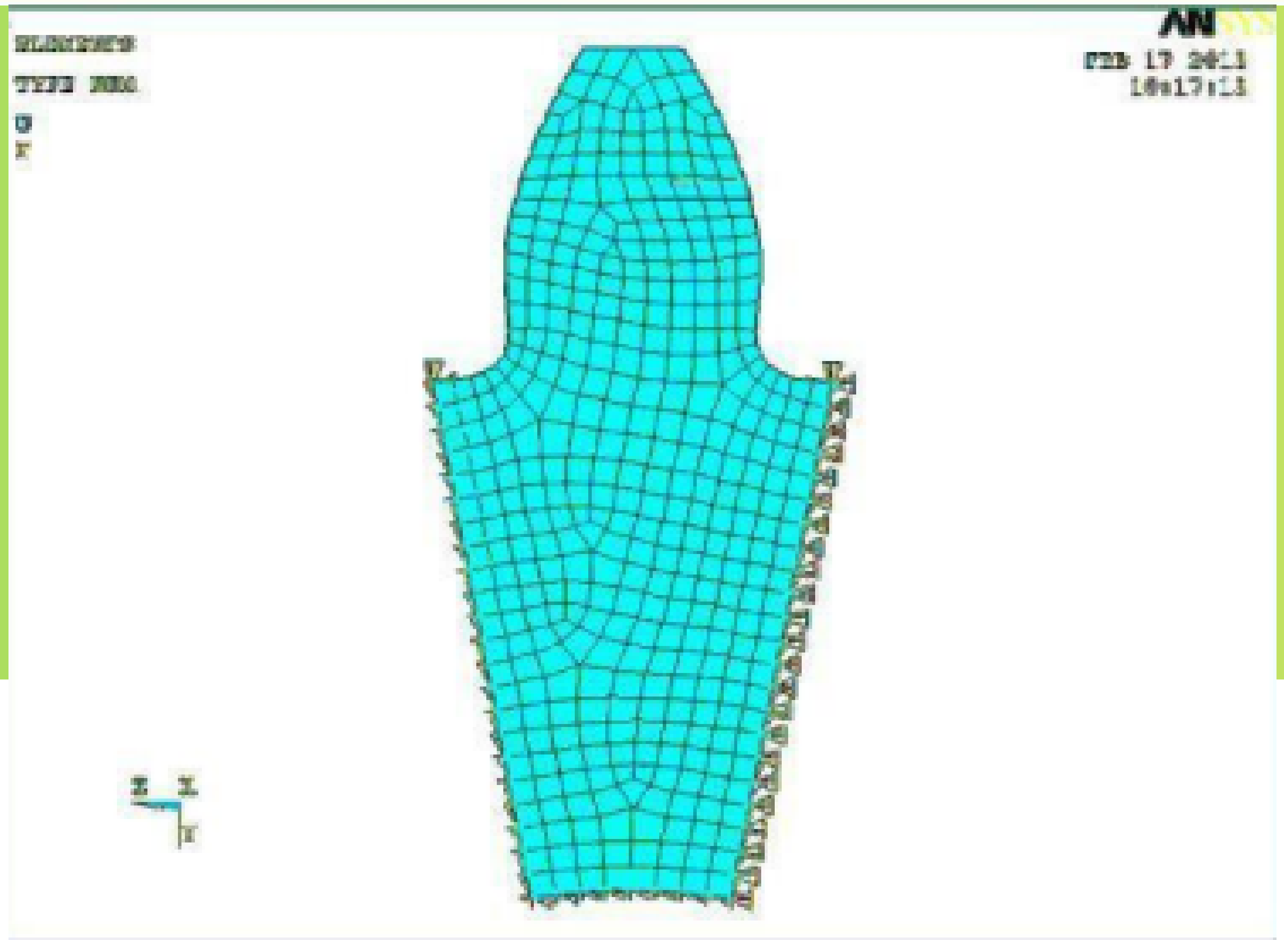
Introduction

- Gears are subject to a lot of bending and contact stresses, reducing life.
- Most are standard designs and not optimized for use under dynamic loading.
- Mostly during custom design, methods like FEA are used which are computationally expensive and also time consuming





Problem Statement

Maximizing Gear Life

Gear Tooth Geometric Parameters

Bending stress and Contact Stress

Importance

- Gear failure is commonly caused by tooth root bending fatigue, surface contact fatigue, Stress concentration at the fillet region.
- Gear failure leads to loss in power transmission, vibration, noise, increased downtime and maintenance.
- Optimizing can help with Longer service life, Higher reliability, Reduced material usage, Improved mechanical efficiency

Reference Model

Name	Value
Centre Distance	100mm
Input Torque	150 N-m
Input Speed	1450 rpm
Material	Steel (grade 1)
Allowable Bending stress	220 MPa
Allowable Contact Stress	900 MPa

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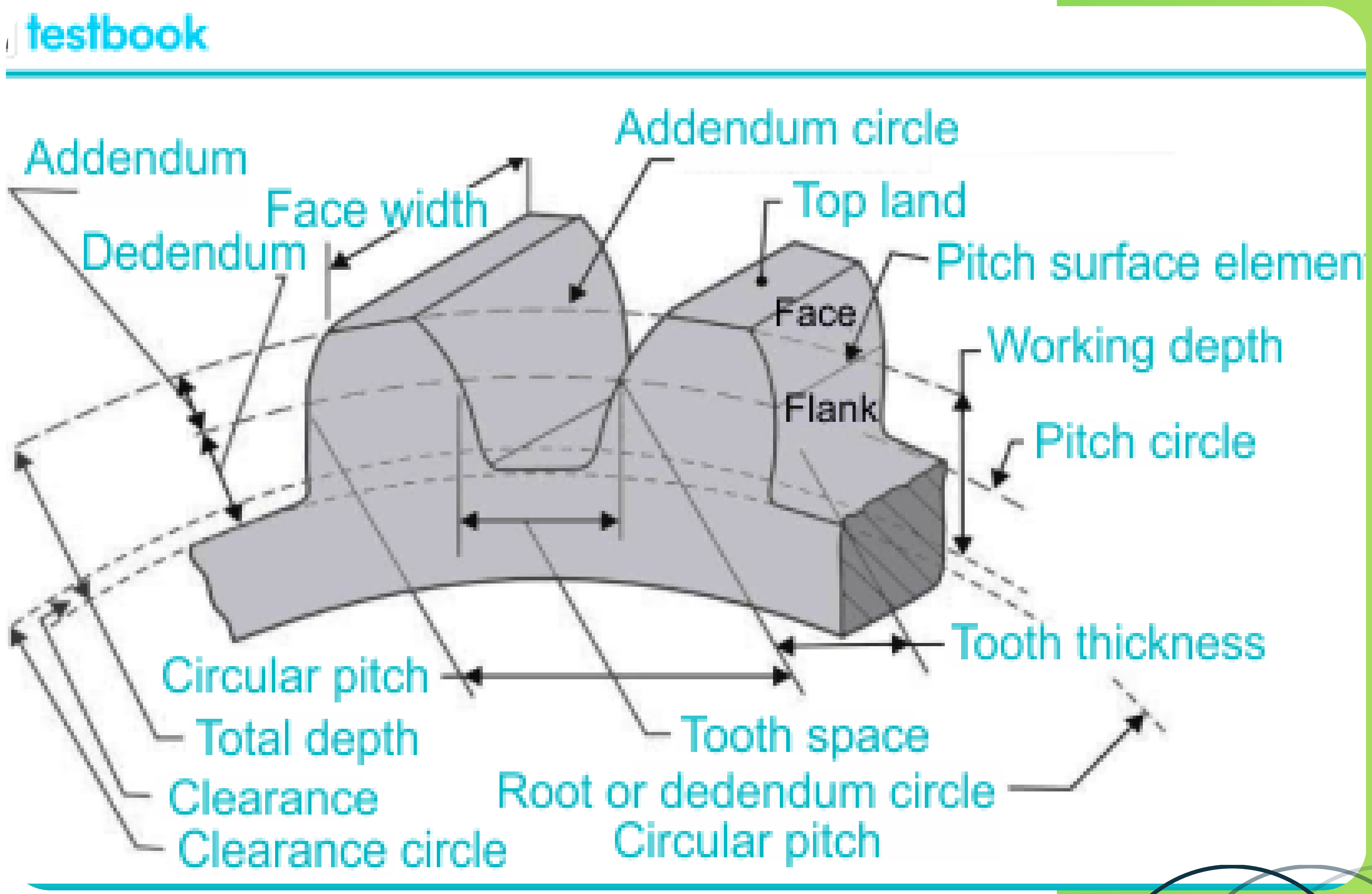
Design Variables

01. Module (m)

02. Face Width (b)

03. Gear Ratio (i)



Constraints

$$a = \frac{m(Z_1 + Z_2)}{2}$$

$$i = \frac{Z_2}{Z_1}$$

$$\varepsilon \geq 1.2$$

$$\sigma_b \leq \sigma_{b,\text{allow}}$$

$$\sigma_c \leq \sigma_{c,\text{allow}}$$

$$8m \leq b \leq 12m$$

$$Z_1 \geq 17$$

Objective Function

$$\text{Life} = \min(N_b,\ N_c)$$

Stress Models

Lewis bending stress Theory

$$\sigma_b = \frac{F_t \cdot K_o \cdot K_v \cdot K_s \cdot K_H}{b \cdot m \cdot J}$$

Hertz contact stress theory

$$\sigma_c = \sqrt{\frac{F_n \left(1 + \frac{r_{p1}}{r_{p2}}\right)}{r_{p1}\, b\, \pi\, C_E \sin\phi}}$$

Failure models

To determine the number of cycles to failure, the study employs S-N based fatigue life equations:

$$N_b = \left(\frac{\sigma_{b,\mathrm{allow}}}{\sigma_b} \right)^{m_{\mathrm{SN}}} \cdot N_0$$

Basquin's Equation (Power-law S-N model): Used to calculate the cycles to failure for bending fatigue.

$$N_c = \left(\frac{\sigma_{c,\mathrm{allow}}}{\sigma_c} \right)^{m_c} \cdot N_0$$

Hertz-based Pitting Model: Used to calculate the cycles to failure for contact fatigue (pitting)

Optimization Procedure

Grid Search

- Compute Z1, Z2, rp1, rp2, V , Ft, Fn.
- Check all constraints
- Compute σb and σc
- Compute Nb and Nc
- Record Life = min(Nb, Nc

Results

$$i : 2.00 \to 1.00\ (-50\%)$$
$$m : 2.00 \to 5.00\ (+150\%)$$
$$b : 20.0 \to 60.0\ (+200\%)$$
$$Z_1 : 33 \to 20$$
$$Z_2 : 66 \to 20$$
$$\varepsilon : 1.737 \to 1.557\ (-10.4\%)$$
$$V : 5.01 \to 7.59\ (+51.5\%)$$
$$K_v : 0.545 \to 0.441\ (-19.0\%)$$
$$J : 0.959 \to 0.799\ (-16.7\%)$$
$$\sigma_b : 129.16 \to 11.05\ (-91.4\%)$$
$$\sigma_c : 1076.13 \to 473.50\ (-56.0\%)$$
$$N_b : 6.45 \times 10^7 \to 3.52 \times 10^{11}$$
$$N_c : 3.42 \times 10^6 \to 4.72 \times 10^8$$
$$\text{Life} : 3.42 \times 10^6 \to 4.72 \times 10^8$$

Thank You!